Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence and Data Science

Name: Krish Makhija	Roll no: 57	Branch: AI&DS
Subject: ASD&D	Batch: T13	Date: 31-07-2026

EXPERIMENT 3: To deploy multi-service applications using Docker Compose

LO Mapped: LO4

LO: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory:

What is Docker?

Docker Compose is a tool used to define, configure, and deploy multi-container Docker applications. It allows multiple services such as web servers, databases, application servers, and caching systems to be managed together using a single YAML configuration file.

Instead of creating and configuring each Docker container separately, Docker Compose uses a file called: dockercompose.yml

This file contains the details of:

- Services to be deployed
- Docker images to use
- Container names
- Ports to expose
- Environment variables
- Storage volumes
- Network connections
- Dependencies between services

Docker Compose is commonly used for applications where multiple containers work together. For example:

- Web server → Nginx/Apache
- Application server → Node.js/Python/Java
- Database → MySQL/PostgreSQL/MongoDB
- Administration tools → phpMyAdmin

Docker Compose YML File Structure:

```
version: '3.8'
services: web:
    image:
nginx
container_na
me:
webserver
    ports:
–"8082:80"

  database:
    image:
mysql:8.0
container_name:
mysqlserver
environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdb
      MYSQL_USER: student
MYSQL_PASSWORD:
student123
    ports:
```

−"3307:3306" volumes:

−mysql-data:/var/lib/mysql

phpmyadmin:

image: phpmyadmin/phpmyadmin container_name: phpmyadmin depends_on: - database

environment:

PMA_HOST: database

MYSQL_ROOT_PASSWORD: root123

ports:

−"8083:80"

app:

image: node:18 container_name: nodeapp

working_dir: /app command: node -e

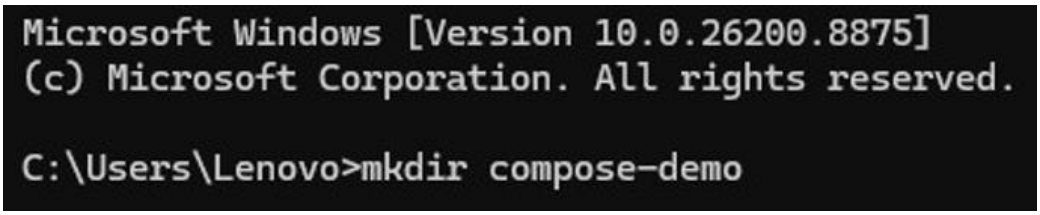
"console.log('Node.js service is running')"

depends_on:

−database
volumes: mysql-data:

Implementation:

1. **Create a project folder**
   mkdir compose-demo



2. Go to the project folder

cd compose-demo

```
C:\Users\Lenovo>cd compose-demo

C:\Users\Lenovo\compose-demo>dir
 Volume in drive C is Windows-SSD
 Volume Serial Number is 9FE2-8C17
```

3. Create the Docker Compose file

4. Verify the file exists

dir

```
C:\Users\Lenovo\compose-demo>dir
 Volume in drive C is Windows-SSD
 Volume Serial Number is 9FE2-8C17

 Directory of C:\Users\Lenovo\compose-demo

07/28/2026  10:36 AM    <DIR>          .
07/28/2026  10:29 AM    <DIR>          ..
07/28/2026  10:36 AM               673 docker-compose.yml
               1 File(s)            673 bytes
               2 Dir(s)  317,759,066,112 bytes free
```

5. Start all services

docker compose up -d

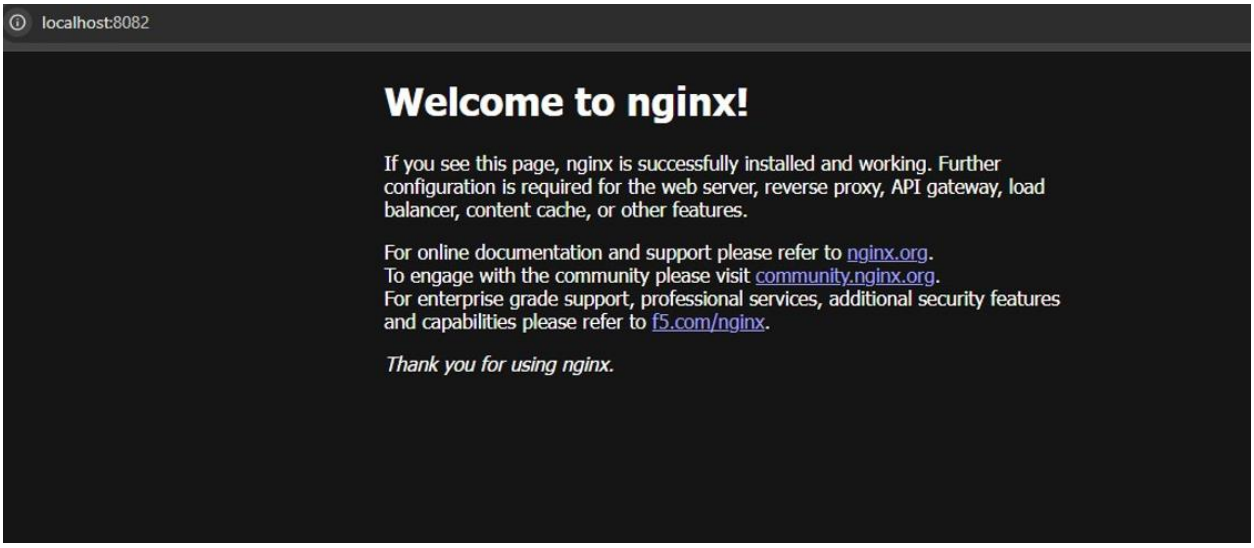
```
C:\Users\Lenovo\compose-demo>docker compose up -d
[+] Running 22/22
 ✔phpmyadmin 21 layers [##########################]      0B/0B      Pulled                                    24.2s
   ✔8c7716127147 Pull complete                                                                               2.5s
   ✔24403a1f6855 Pull complete                                                                               0.6s
   ✔e1cf44d6017a Pull complete                                                                               5.6s
   ✔2489d5e860a7 Pull complete                                                                               2.3s
   ✔0248257cbd51 Pull complete                                                                               3.1s
   ✔dd53cf9bf4cf Pull complete                                                                               3.1s
   ✔a139c2f3234a Pull complete                                                                               3.7s
   ✔6571cfdbe5b2 Pull complete                                                                               5.5s
   ✔8d83c968ca9a Pull complete                                                                               4.4s
   ✔fddb92e888a7 Pull complete                                                                               5.9s
   ✔749b92ea0995 Pull complete                                                                               6.1s
   ✔4eed3454c20c Pull complete                                                                               6.2s
   ✔00ef78e422f0 Pull complete                                                                               6.6s
   ✔004f06ab2f6c Pull complete                                                                               6.7s
   ✔4f4fb700ef54 Pull complete                                                                               6.9s
   ✔273a09302c4a Pull complete                                                                               8.3s
   ✔a8c81cca4b8b Pull complete                                                                               7.3s
   ✔271012172ea0 Pull complete                                                                               9.0s
   ✔0bebf457381d Pull complete                                                                               7.9s
   ✔30b083700c69 Pull complete                                                                               8.6s
   ✔b5a8243c1a5e Pull complete                                                                               9.9s
[+] Running 0/5
 - Network compose-demo_default      Created                                                                  0.8s
 - Volume "compose-demo_mysql-data"  Created                                                                  0.7s
 - Container webserver               Starting                                                                 0.7s
 - Container mysqlserver             Starting                                                                 0.7s
 - Container phpmyadmin              Created                                                                  0.2s
```

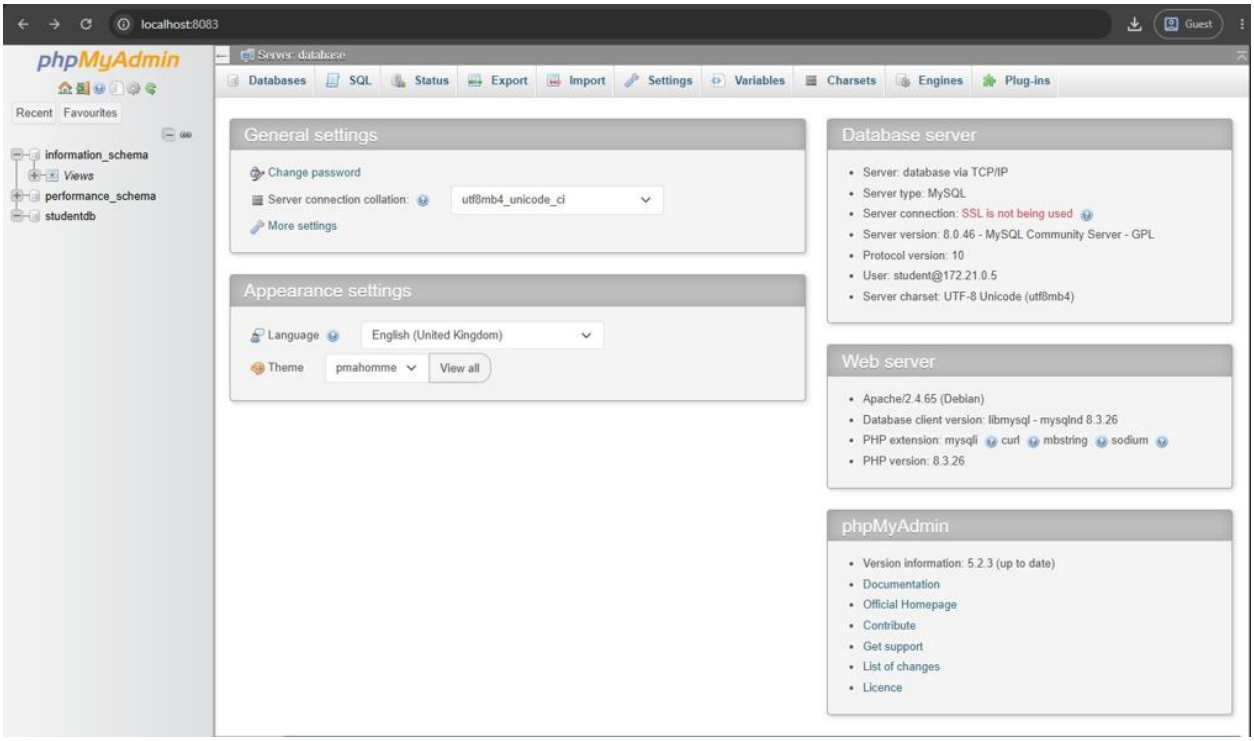
6. Check whether the containers are running

docker ps

```
C:\Users\Lenovo\compose-demo>docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS          PORTS                                 NAMES
cba4003dcd63   phpmyadmin/phpmyadmin  "/docker-entrypoint.…"   10 seconds ago   Up 8 seconds    0.0.0.0:8083->80/tcp                  phpmyadmin
78b670039593   nginx                  "/docker-entrypoint.…"   10 seconds ago   Up 8 seconds    0.0.0.0:8082->80/tcp                  webserver
38f4a5c05836   mysql:8.0              "docker-entrypoint.s…"   3 minutes ago    Up 3 minutes    33060/tcp, 0.0.0.0:3307->3306/tcp     mysqlserver
```

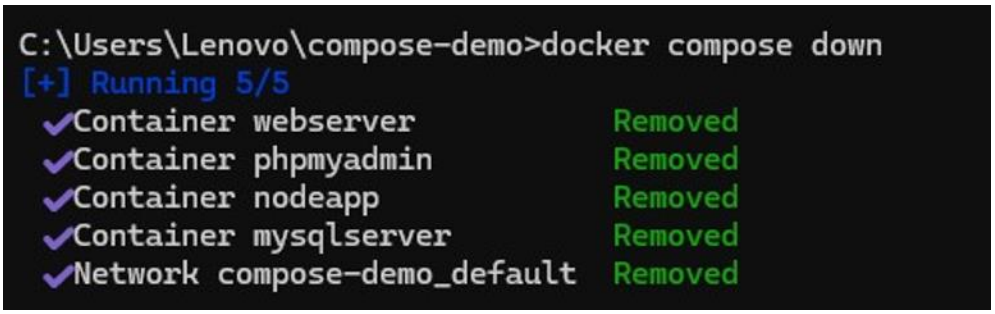
7. Test the services





9. Remove the containers

docker compose down



Conclusion:

A multi-service application was successfully deployed using Docker Compose. Multiple containers such as web servers, databases, and application services were configured, created, and managed using a single dockercompose.yml file and Docker Compose commands.